

6 Fig 6.1 shows an electron diffraction tube.

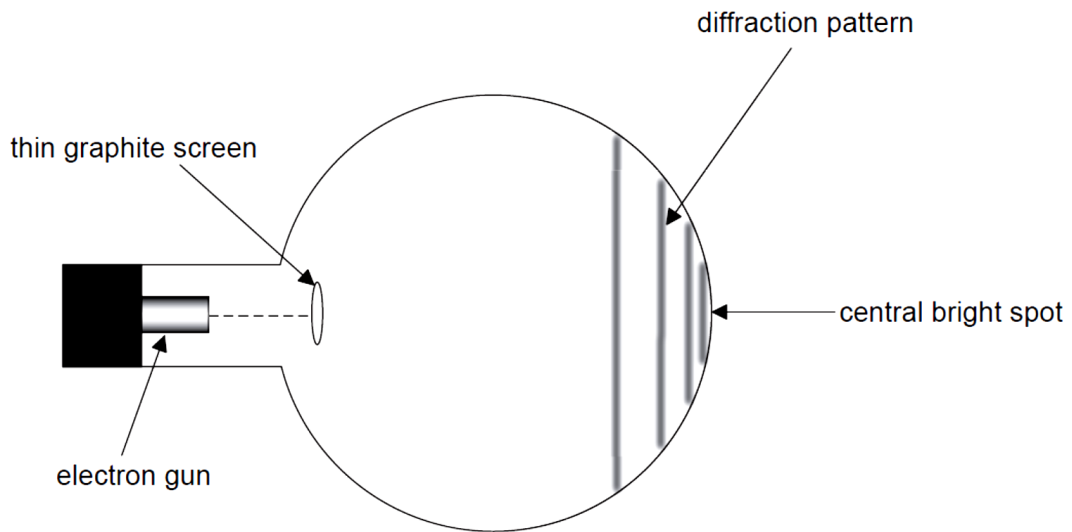


Fig 6.1

A diffraction pattern of concentric circles is formed when diffracted electrons are incident on a fluorescent layer at the end of the tube.

(a) (i) Explain the meaning of the term *diffraction*.

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 [1]

(ii) State the condition necessary for electrons to produce observable diffraction pattern when passing through matter, such as a thin sheet of graphite.

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 [1]

(b) An electron with charge $-q$ and mass m is made to accelerate from rest in the above evacuated tube through a potential difference of $V = 120 \text{ V}$.

(i) State what is meant by the *de Broglie wavelength*.

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 [1]

- (ii) Calculate the de Broglie wavelength of the electron.

de Broglie wavelength = m [4]

- (iii) The potential difference V is then increased slightly above 120 V. State and explain how the diffraction pattern formed on the fluorescent layer would change.

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..... [3]

[Total: 10]